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Designing a Consensus Mechanism for Blockchain Cross-chain Applications

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Final decision: REJECT

The paper is AI-generated and contains in many parts what is usually called AI-hallucinations; the respected authors probably do not know anything about it.

Designing a Consensus Mechanism for Blockchain Cross-chain Applications

[redacted]

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When tracing through the paper what the authors exactly means by this,

It's already done by Polkadot !!!

Keywords: blockchain cross-chain; online community; knowledge sharing; WNPoS consensus mechanism

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1 Introduction

Blockchain is a decentralized and distributed ledger technology that employs peer-to-peer communication, encryption algorithms, consensus mechanisms, and additional technologies to organize transaction records into data blocks that are globally consistent, tamper-proof, and unforgeable. These blocks are ~~then~~ sequentially linked in chronological order to create a storage structure [1–3]. Given the vast potential of blockchain, numerous projects have emerged, resulting in the establishment of various blockchain infrastructures and business systems [4]. As these blockchains are applied, there is a growing demand for data exchange, value circulation, and collaboration, emphasizing the necessity for cross-chain interactions [5–6]. For instance, during knowledge sharing within blockchain-based online communities, multiple entities,(such as users, communities, identity authentication centers, and copyright management organizations,) exist independently yet remain interconnected, creating a significant demand for cross-chain capabilities. As users share and acquire knowledge through online communities, they also seek robust copyright protection for their intellectual contributions [7]. Traditional copyright protection measures based on blockchain typically rely on single-chain or consortium blockchains, where the copyright protection agency functions as a single node on the chain. This model restricts the agency's ability

the word "then" could imply blocks are linked afterwards. So, better remove it or write the phrase if you prefer.

which emphasizes (to not mix so many ",")

instead of ","

This sentence is about another example where cross chain capabilities are needed, so either start with "Also" or better put each ex. in a new paragraph/bullets/numbering. Better put Finance or trading here too, since it is the most important (first existed) motive for this research

is this the right way to describe it? "a single node in the chain"? let me check fisrt

or "cannot handle copyright violations through other chains"?

Are you talking about this node efficiency in other operations (as a validator for ex.) or do you mean the above?

to process only transactions within its own chain, thereby impacting its efficiency in handling other operations. In contrast to single-chain or consortium blockchains, cross-chain solutions offer greater advantages for knowledge sharing and intellectual property protection, such as enhanced transaction throughput, quicker processing speeds, and improved operational efficiency within a specified timeframe [8–9].

ref [47] belongs here

However, in complex cross-chain systems, the consensus mechanism is vital for ensuring normal, stable, and efficient operation [10–11]. Existing consensus mechanisms exhibit certain limitations when implemented in cross-chain environments. Thus, researching more appropriate consensus mechanisms tailored for blockchain cross-chain systems is of significant importance, as highlighted in the following aspects.

First, in terms of theory:

(1) It can drive innovation in consensus mechanisms. Consensus mechanisms are a core component of blockchain technology. Research on consensus mechanisms based on blockchain cross-chains can contribute to the further development of blockchain technology, particularly in enhancing system performance, security, and scalability.

(2) It helps enrich blockchain governance theories. Blockchain governance is crucial for ensuring long-term stable operation and continuous innovation. Exploring the establishment of a fairer, more transparent, and efficient governance mechanism provides new perspectives and references for the development of blockchain governance theories.

Second, in terms of practice:

(1) It helps broaden application scenarios. The application of cross-chain consensus mechanisms is not limited to the aforementioned field of knowledge sharing in online communities but also extends to various sectors, such as finance, supply chain management, and the Internet of Things, promoting the widespread application and in-depth development of blockchain technology in these areas.

better to elaborate a little above

(2) It aids in the healthy development of the blockchain ecosystem. A well-structured consensus mechanism can strengthen the decentralization of blockchain cross-chain systems, facilitate fair participation among diverse roles, and support the sustainable growth of the blockchain ecosystem.

2 Related Research

2.1 Existing mainstream consensus mechanisms

In a P2P network, the process where mutually untrusted nodes ultimately achieve data consistency by following a predefined mechanism is known as consensus. The consensus algorithm is a core mechanism in blockchain systems. The most commonly used consensus algorithms in

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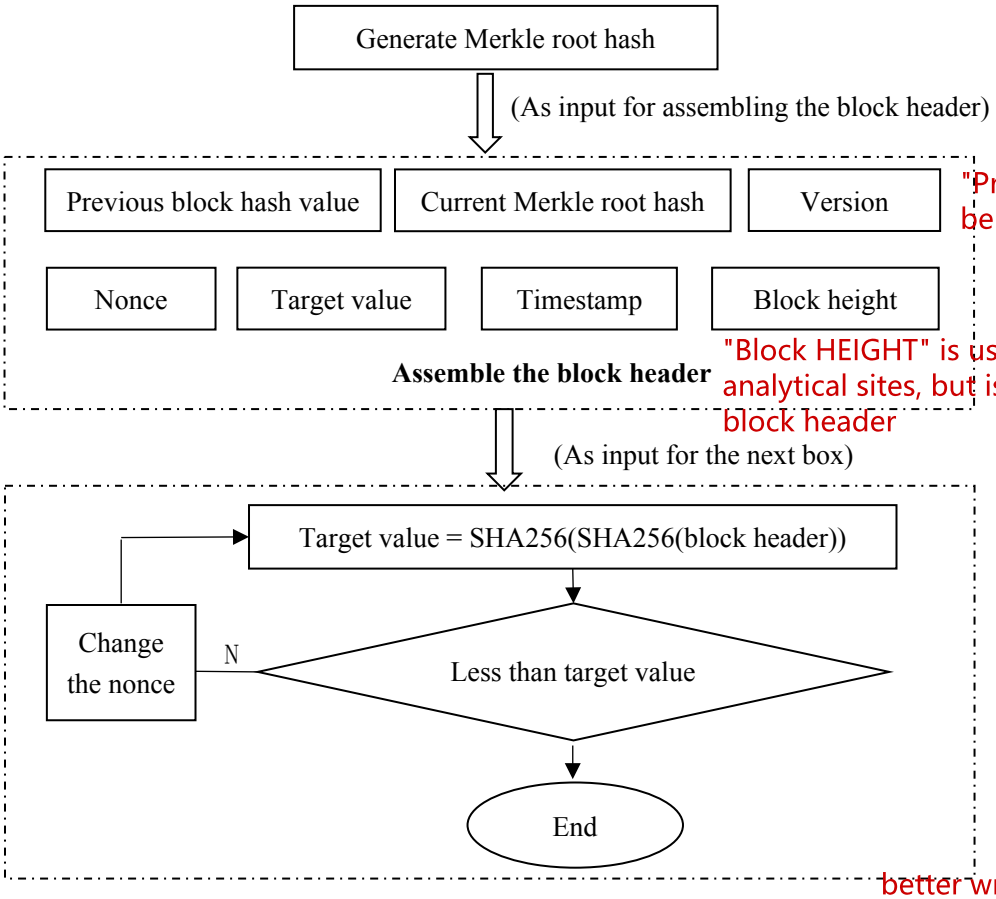
mainstream blockchain systems include proof of work (PoW) [12–13], proof of stake (PoS) [110], and Byzantine fault tolerance (BFT) algorithms.

PoW is the consensus algorithm utilized by the Bitcoin system. In this system, a mechanism is established where nodes compete for the right to record transactions based on their computational power, or "hashing power." The process consists of the following steps. First, blockchain nodes gather a list of recently generated transactions that have not yet been added to the blockchain and apply the Merkle tree algorithm to produce the root hash value of these lists, which serves as the input for the entire algorithm. Second, this value is combined with additional fields, such as the timestamp, version number, nonce, and target hash value, to create the block header. Finally, the nonce in the block header is continuously modified, and a double hash operation is conducted on the new block header. The result is then compared to the target value in the block header; if it is smaller, the node secures the right to record the transactions, concluding the algorithm. This process can be visually illustrated in Figure 1.

and wins the block reward, you mentioned it in POS and should mention it here too.



"Protocol" version to be more clear

"Block HEIGHT" is usually added by analytical sites, but is NOT part of the block header

Figure 1 Process of the proof of work (PoW) consensus mechanism

better write in Bitcoin as an example POW consensus blockchain

The benefits of the PoW consensus mechanism are evident in its strong decentralization features and its capacity to maintain system security. In the Bitcoin framework, an attacker would require a minimum of 51% of the network's computational power to alter data on the main chain,

and the substantial computational expenses guarantee the secure and stable operation of Bitcoin [14]. Nevertheless, the PoW consensus mechanism also presents clear disadvantages: First, the ongoing expenditure of computational power to achieve the target value leads to considerable resource wastage, second, the efficiency of transactions is limited. The current throughput of the Bitcoin network is 7 transactions per second, which represents a significant shortfall compared to other leading blockchains; third, the confirmation time for Bitcoin transactions can extend up to one hour, rendering Bitcoin inappropriate for transactions requiring quick processing [15].

you may add before the semicolon "and large carbon footprint"

this is not "third" it is also about the timing factor (which is the real third disadvantage). the 2nd about TX should talk about the limited functionality of TX (compared to coding capabilities in smart contracts)

In response to the challenges associated with PoW, researchers have made efforts to develop new consensus algorithms to enhance blockchain functionality, with the equity-based consensus algorithm PoS being the most notable. Unlike PoW, where all nodes are eligible to compete for the right to record transactions, PoS permits only those nodes that have staked a specified number of tokens to engage in the competition [16–17]. Based on this framework, an appropriate algorithm is employed to select the node entitled to record transactions from among those participating. This selection process is designed to be scientific and objective, ensuring results that are neither predictable nor manipulable. ~~Specifically~~, the competition rule for acquiring the right to record transactions in PoS stipulates that the greater the coin age (coin age = number of tokens * duration of token holding), the higher the likelihood that a node will secure the right to generate a block. The node awarded the right to record transactions packages transaction records into a block and disseminates it to the blockchain through a network broadcast, subsequently receiving a corresponding token reward [18–19]. Compared to the PoW consensus mechanism, PoS offers several advantages. First, PoS demonstrates greater energy efficiency by basing the competition for transaction recording on ~~equity,~~ stakes thus reducing the time needed for consensus across the network and minimizing resource wastage inherent in the PoW approach. Second, PoS promotes equity, as coin age correlates positively with both the number of tokens and the duration of token holding. Consequently, candidate verification nodes with fewer tokens but longer holding periods can also be selected as block producers, enabling "senior minority shareholders" to compete with "young majority shareholders." Following the publication of a block by a block-producing node, its coin age is reset, which mitigates the issue of concentrated computational power found in the PoW mechanism. Third, the PoS consensus mechanism offers an annual interest rate on staked tokens, generating additional interest. Thus, the currency in circulation within the system is not solely derived from mining, and the dual methods of currency generation significantly raise the barriers to a "51%" attack, thereby enhancing system security [20]. However, the PoS consensus mechanism is not without its challenges [21]. For instance, nodes holding equity may lack the intention to participate in the competition for block production but are still required to do so; additionally, offline

this is an example rule, does not apply to all POS blockchains (it is the rule of Peercoin, yes the first but not the most famous or most common to generalize the talk according to it).

again, this is not a general case advantage

you may move all of this under Peercoin case example as the first POS; ie., telling the story in historical order.

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nodes continue to accumulate coin age, which may not accurately represent their engagement.

the 1st POS suggested consensus rule

In response to the aforementioned issues with PoS, the delegated proof of stake (DPoS) algorithm was introduced to enhance the engagement and accountability of token holders in mining activities [22]. The primary mechanism of the DPoS consensus is to select a limited number of representatives from all eligible candidates for the purpose of record-keeping, with candidates determining the specific representatives. These elected representatives are required to remain online for extended durations and must stake a portion of their tokens as a deposit to ensure the fulfillment of their responsibilities. The representatives take turns creating new blocks and receive a percentage of the transaction fees associated with each block, incentivizing them to actively contribute to the security of the system. Additionally, all actions performed by the elected representatives, including packaging and verification, are monitored by other nodes. If a representative node fails to carry out its duties adequately, its staked tokens may be confiscated, and its representative status can be revoked. Within the DPoS consensus mechanism, the relatively small number of nodes competing for the right to record transactions leads to a marked increase in the online presence of nodes and enhances the speed of block validation and generation [23–24]. Nonetheless, DPoS is not without its limitations, such as the potential for corruption during the selection of delegate representatives, a small number of authorized delegates, and the risk of collusion among multiple delegates.

The BFT consensus mechanism is distinct from the previously mentioned PoW, PoS, and DPoS mechanisms in that it does not involve the selection of block-producing nodes. Rather, it depends on direct voting by network participants to determine the creation of new blocks [25]. When participants reach consensus, the new block is added to the blockchain. Consistency is a core requirement of the BFT consensus mechanism. As outlined by the CAP theorem [26], a distributed system cannot simultaneously achieve consistency, availability, and partition tolerance. Therefore, for the BFT consensus mechanism to meet the "availability" requirement while maintaining "consistency," it must compromise on "partition tolerance." Furthermore, the BFT consensus mechanism requires mutual understanding among system participants, which is why it is primarily utilized in consortium or private blockchains.

most BFT varients involve a leader election process, and exists from distributed systems (before blockchains) it is just do not necessitates coupling block production with block broadcasting. I'd say the main feature is a defined no. of nodes, that's why in consortium or after committee selection in POS/DPOS

The original BFT consensus mechanism is constrained in practicality due to its reliance on supplementary synchronization mechanisms, resulting in an algorithmic complexity that increases exponentially with the number of nodes. The practical Byzantine fault tolerance (PBFT) algorithm alleviates this problem by decreasing the complexity of Byzantine agreement protocols from exponential to polynomial, thereby enabling their application in distributed systems [27]. PBFT functions as a state machine Byzantine system [28], requiring all nodes to behave consistently. This consistency is ensured through three core protocols: the consensus protocol, the checkpoint protocol,

I'd say you rewrite this section where Peercoin is mentioned as the 1st POS after Bitcoin, then Cardano as the DPOS you are describing (although slashing as punishment is in Ethereum), then mention BFT as brought from traditional systems to handle committee and also private closed blockchains (which seems to be the targeted by this paper)

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and the view change protocol. The consensus protocol stipulates that service nodes must process client requests in a specified sequence. The server is composed of one primary node and multiple replica nodes, where the primary node is tasked with ordering client requests, and the replica nodes execute these requests in the established order. All server nodes operate under uniform configuration information, known as a view, which changes upon the replacement of the primary node. The consensus protocol involves several stages, including Request, Pre-prepare, Reply, Prepare (mutual interaction), and Commit. The operational mechanism of the PBFT consensus is illustrated in Figure 2 [29–31].

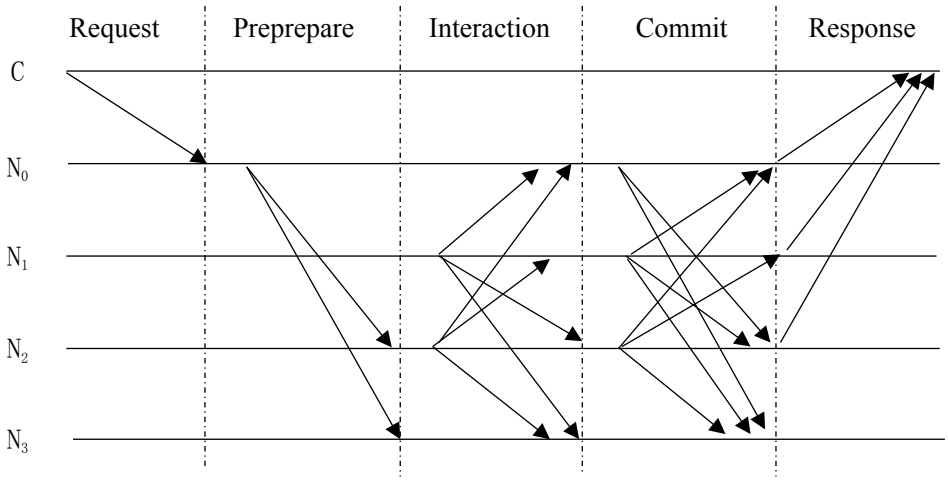

Figure 2 PBFT Algorithm Flowchart

In general, PBFT is more suitable for private or consortium blockchains that require strong consistency. For instance, in the Hyperledger blockchain, PBFT is one of the optional consensus protocols.

All the consensus mechanisms have their respective strengths and weaknesses. A comparison of the advantages and disadvantages of the aforementioned mainstream consensus mechanisms [32–34] is presented in Table 1.

Table 1 Comparison of mainstream consensus mechanisms

Consensus Mechanism	Fault Tolerance	Blockchain	Advantages	Disadvantages
PoW	50%	Public Blockchain	Good decentralization and high security	Low transaction efficiency and waste of significant computing power
PoS	< 1/3 = 33% ~~50%~~	Public Blockchain	Energy-saving, environmentally friendly, and high-performance	May lead to a monopoly of computing power
DPoS	< 1/3 = 33% ~~50%~~	Public Blockchain	Resource-saving and high consensus efficiency	Lower security and complex implementation
PBFT	< 1/3 = 33% ~~30%~~	Federated Blockchain/Private Blockchain	Strong security and stability, and high consensus efficiency	Not suitable for public blockchains

since most POS (even DPOS) uses a variant of BFT after committee selection, they share the same threshold of <1/3 (sometimes they try to enhance on that by classifying nodes into different categories or other ways, but as a general ratio for POS you should write < 1/3)

In addition to the conventional consensus mechanisms previously described, numerous new algorithms have been implemented in blockchain systems. For instance, to mitigate the limitations of the proof-of-work (PoW) consensus mechanism, including reduced transaction efficiency and prolonged confirmation times, Sompolinsky [35] introduced the GHOST-PoW consensus mechanism. By leveraging the benefits of proof of stake (PoS), Byzantine fault tolerance (BFT), and verifiable random functions (VRF)*, the ontology blockchain employs a verifiable random function-based Byzantine fault tolerance (VBFT) consensus mechanism, facilitating rapid network consensus [36]. Huang [37] put forth the RBFT consensus algorithm, which incorporates the Raft mechanism to designate a leader responsible for forming a committee to achieve PBFT consensus. Wang [38] developed the OBFT consensus algorithm, which addresses leader selection and communication delays associated with PBFT to some degree. To enhance the efficiency and security of the PBFT consensus algorithm, Wang Rihong [39] proposed a more effective supervisory Byzantine fault tolerance algorithm. Hou Kaixiang et al. [40] presented a dynamic blockchain consensus algorithm with preverification (DBCP) to resolve challenges such as high overhead and decreased credibility. Li Fengqi et al. [41], recognizing the potential for malicious behavior across various terminals in the Industrial Internet of Things, introduced an efficient consensus algorithm

I prefer you rewrite this paragraph in general, elaborating that all those papers suggest different ways for leader election (the block proposer) and then applies PBFT as you stated.

Note that leader election is important because if predictable could be hacked to cause a Dos attack, or bribed to put your transactions first.

* you should clarify this term when it is first introduced in the paper; since it is not heavily used in the paper, you may put in a footnote: explain that it provides randomness to make the leader unpredictable and try to add a link for further details.

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based on credit scoring and double-layer dynamic practical Byzantine fault tolerance (CD-PBFT) to fulfill the requirements for high efficiency, security, and fault tolerance during the consensus process, among other strategies.

2.2 Review of Research

The above research reveals that the generation and packaging of blockchain transactions need to follow corresponding rules and that the consensus mechanism plays a vital role in this process. It is a core component of the blockchain system, ensuring its stable operation. A good consensus mechanism can enhance the system's security, scalability, throughput, and other performance metrics, thereby promoting the widespread application of blockchain technology [42].

Again, relays are not the only way to handle cross-chain communication in blockchains, you discuss a special case in a paper as if it is the general theme

In blockchain-based cross-chain systems, a relay chain-parallel chain architecture is employed as the foundational structure, which involves a relay chain along with multiple parallel chains [43]. Complex cross-chain systems necessitate robust consensus mechanisms to facilitate their efficient operation. First, additional roles are required to maintain the normal functioning of the system. In consensus mechanisms such as ~~PoW~~, PoS, and PBFT, the primary role is that of the validator node, which complicates the coordinated and stable operation of various chains within a complex cross-chain system. Besides validator nodes, other management roles must be incorporated into cross-chain systems. Second, operational efficiency is critical to circumvent the limitations of existing consensus mechanisms. Among traditional consensus mechanisms, PoW is characterized by low transaction efficiency and significant waste of computing resources; PoS can lead to a concentration of computing power; and PBFT is ill-suited for public blockchains, among other challenges. These issues highlight the necessity for identifying more appropriate consensus mechanisms for cross-chain systems. Finally, the interests of all participants should be safeguarded to the greatest extent possible. Given the involvement of numerous entities in cross-chain systems, ensuring the overall stability of the system requires the protection of all participants' interests as much as feasible.

you haven't identified the term ""validator" yet, again better introduced in its proper place in sec 2.1, not in POW by the way.

You should first clarify the main problem as that the 2 communicating blockchains could have completely different types and/or consensus mechanisms, and the cross chain must handle (understand both) then have its own consensus mechanism to pass relay messages according to it. Also, TX time=time in 1st +cross-chain time +time in 2nd chain -security/FT = the least (weakest) in them

To ensure the reliable, secure, and efficient functioning of blockchain cross-chain systems, there exists an urgent requirement for novel consensus mechanisms.

3 Design of the WNPoS Consensus Mechanism

3.1 Introduction of the NPoS Consensus Mechanism

Based on the analysis presented, this paper introduces a consensus mechanism termed weighted nomination proof of stake (WNPoS), which is inspired by the consensus mechanisms utilized in blockchain platforms such as Polkadot and Ethereum [44–46]. WNPoS is a relative concept to PoS (Proof of Stake) and NPoS (Nominated Proof of Stake). In the PoS consensus mechanism, a participant's likelihood of being granted the right to record a new block increases with the number of tokens (stakes) held, resulting in a scenario where wealth accumulates among the

Refs [44-45] has nothing to do with the issue, even [46] is not accurate (Polkadot has 2 protocols papers and an html file as summary). + Add an Ethereum reference

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already affluent. Conversely, the NPoS consensus mechanism mitigates this issue by incorporating (both have multiple validator nodes) ~~multiple validator nodes within the blockchain system~~. n Nominator nodes that stake their tokens to nominate validator nodes they trust, with all chosen validator nodes receiving equal rewards. Consequently, nominator nodes tend to select validator nodes that have a higher probability of the main point is that they trust they will be active and not malicious not necessarily, or is this your proposed strategy? election but lower overall staking amounts, in an effort to optimize their own rewards. This NPoS approach takes into account the interests of the "minority," thereby promoting enhanced fairness.

In the Polkadot site: Nominators can choose to backselect validators with their stake. However, the NPoS mechanism treats the weights of all nominator nodes equally during the nomination process, leading to a potential inconsistency. A more rational approach would involve recognizing that different users have varying degrees of nomination power, similar to shareholders with more shares possessing greater voting rights in corporate elections. In this regard, nominator nodes should have distinct weights based on the number of tokens held, with those holding larger amounts carrying greater significance than those with fewer tokens. Therefore, this paper proposes the integration of weights into the nomination process within the NPoS consensus mechanism, culminating in the "Weighted Nomination Proof of Stake" consensus mechanism. NOT TRUE

In the subsequent sections, the participating roles and algorithms of the WNPoS consensus mechanism will be designed, followed by an analysis of its characteristics.

3.2 Roles in the WNPoS Consensus Mechanism

The participants in the WNPoS consensus mechanism include validator nodes, collector nodes, nominator nodes, and supervisor nodes.

Validator nodes are tasked with validating and confirming the integrity of blocks within the blockchain system. These nodes operate in both the parallel chains and the relay chain and are selected by other nodes based on nominations and specific criteria. High availability and bandwidth are required for validator nodes to ensure constant online status while verifying and confirming newly generated blocks on parallel chains. This process entails receiving, validating, and publishing candidate blocks. Multiple validator nodes are present on both the relay chain and each parallel chain, forming a set of validator nodes. This set collaborates to effectively approve new blocks on their respective parallel chains, updating transaction queue states, transferring data from the output queue of one parallel chain to the input queue of another, processing confirmed transaction batches from the relay chain, and auditing final blocks. Failure to fulfill these responsibilities may result in the forfeiture of all or part of the tokens staked by the validator nodes.

how would you measure that? if you come up of a new methodology, maybe it could lead to a new protocol but Weighted voting power based on tokens is Polkadot

Collector nodes are tasked with generating valid blocks for validator nodes on parallel chains while observing network conditions and supplying information to validator nodes regarding nodes that demonstrate malicious behavior. Collector nodes aid validator nodes in producing valid blocks for the parallel chain by operating a full node dedicated to a specific parallel chain. They retain all

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essential information required for the authorization of new blocks, the packaging of new blocks, and the execution of transactions, akin to the role of miners in proof-of-work (PoW) consensus-based blockchains. During normal operations, collector nodes gather and execute transactions, formulate an "unsealed" block, and submit it along with validity proof information to one or more validator nodes that are currently responsible for evaluating that parallel chain's block.

Nominator nodes engage in the selection of one or more reliable validator node candidates through the stacking of tokens. By electing validator nodes, nominator nodes assign the duty of network maintenance to these validators on their behalf. Based on their token contributions, rewards and penalties are allocated to nominator nodes in proportion to their share of the total stake possessed by validator nodes.

Supervisor nodes oversee various activities across the platform to identify and prevent malicious behavior by other participants. Similar to full nodes in blockchain systems, supervisor nodes necessitate relatively fewer resources and are not required to maintain stable uptime and bandwidth; however, a deposit must be bound to ensure proper functioning.

The engagement of various roles—validator nodes, collector nodes, nominator nodes, and supervisor nodes—ensures the stable functioning of the cross-chain system within the blockchain's online community. The illustration of these nodes is provided in Figure 3.

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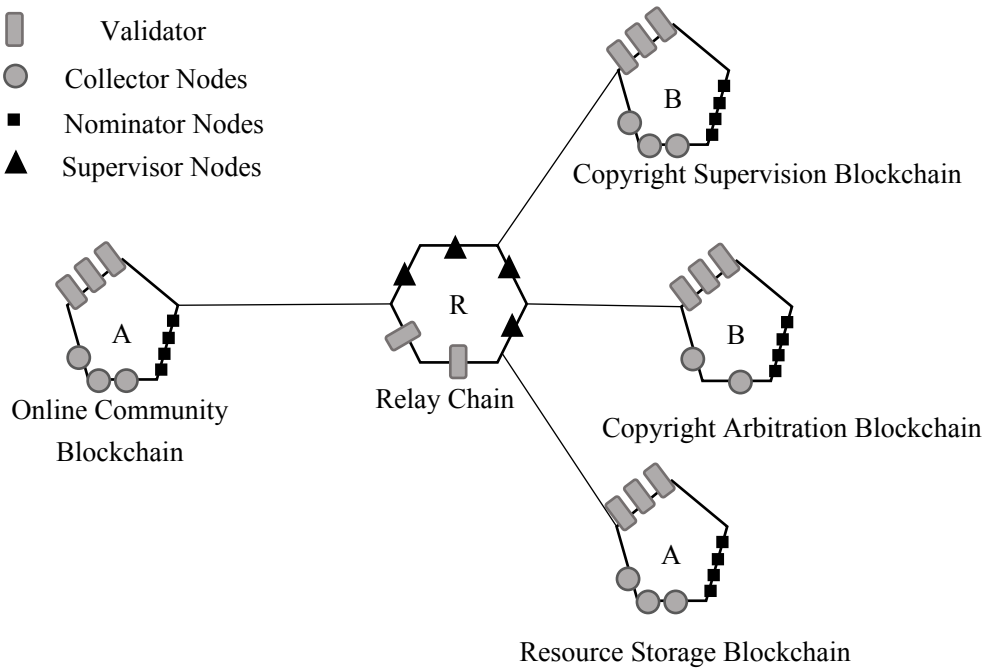

Figure 3: Existence Forms of Nodes in Consensus Mechanisms

In the aforementioned blockchain system, collector nodes observe the block generation process and the consensus protocol of parallel chains in real time, functioning as full nodes. Through this

observation, the candidate block, which is the most probable block to become the next on the parallel chain, is identified.

Once the data created by this candidate block are signed, the collector node forwards the information to validator nodes nominated by nominator nodes within the same parallel chain. These validator nodes assess the legitimacy and validity of the data in the received candidate block. If the verification process reveals no issues, the block is officially generated; if any issues are detected, its generation is denied.

During the block generation process by validator nodes, supervisor nodes oversee the specified parallel chain for any violations, such as the publication of unqualified blocks. If such violations are detected, malicious behavior is reported to other validator nodes for reverification.

The interaction of information among these roles is depicted in Figure 4 [47].

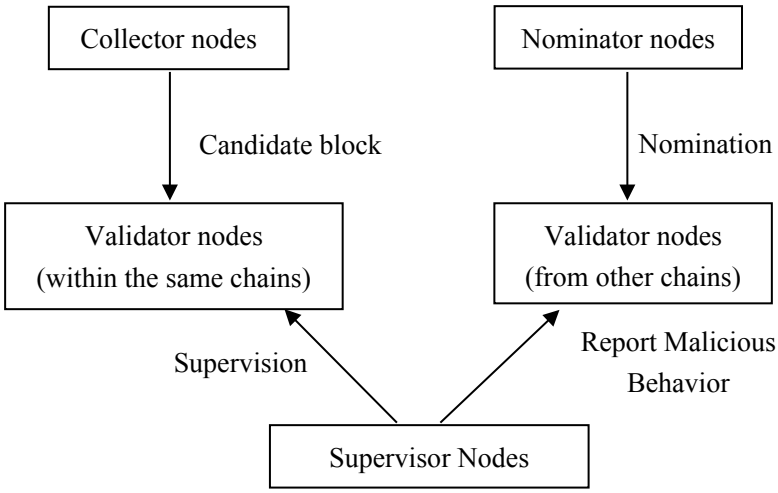

Figure 4 Information interactions among the four roles in the consensus mechanism

3.3 WNPoS Consensus Mechanism Algorithm

In the blockchain consensus mechanism, the creation and functioning of validator nodes (hereinafter referred to as validator nodes for both formal and candidate validator nodes) represent the most critical elements. The "weighting" within the weighted denominated proof-of-stake (WNPoS) consensus mechanism is primarily reflected in the process of generating validator nodes.

In a knowledge-sharing blockchain system, any node that holds tokens can act as a nominator node. During each cycle (e.g., 24 h), nominator nodes select to stake their tokens in support of one or more candidate validator nodes of their choice. The amount of tokens staked by nominator nodes for each candidate validator node may vary. Candidates that attract more staked tokens have an increased likelihood of being appointed as formal validator nodes. Therefore, determining validator nodes involves calculating the maximum stake of candidate validator nodes. Given that validator nodes are generated in rounds, with an assumption of 10 validator nodes for the relay chain or each

parallel chain, a total of 10 rounds of elections is necessary, with one validator node elected in each round. Additionally, because the stake amount of candidate validator nodes is inversely proportional to the nomination cost incurred by nominator nodes, the problem of calculating the maximum stake of candidate validator nodes can be reframed as minimizing the nomination cost for nominator nodes. This challenge can be resolved using a "greedy algorithm." To elucidate the algorithm's procedure, the term "load" is introduced, which refers to the cost incurred by a nominator node when nominating a specific validator node, denoted as $load(N_i)$ for the load of the *i-th* nominator node.

Assuming that T represents the set of nominator nodes, Y represents the set of candidate validator nodes, $N_{i\in T}$ denotes the *i-th* nominator node, $\mathrm{stake}(\mathrm{N}_{i\in T})$ represents the weight (stake) of the *i-th* nominator node, $V_{j\in \mathrm{Y}}$ is the *j-th* candidate validator node, $approval_stake(V_{j\in Y})$ is the total weight supporting the *j-th* candidate validator node, and $value(V_{\mathrm{j}\in \mathrm{Y}})$ represents the minimum load value when selecting V_{j} as a formal validator node. The rules for the WNPoS validator node generation algorithm are as follows:

(1) Create a list of all nominator node elections, including the weights (stakes) generated on the basis of their total staked tokens and the validator nodes they support.

(2) Calculate the total weight supported by each candidate validator node on the basis of the weights of the nominator nodes, that is

$$approval_stake(V_{j\in \mathrm{Y}}) = \sum_{i\in T} \mathrm{stake}(\mathrm{N}_{\mathrm{i}} \rightarrow V_j) \qquad \text{(Formula 1)}$$

where $\mathrm{stake}(\mathrm{N}_{\mathrm{i}} \rightarrow V_j)$ represents the support weight of the *i-th* nominator node for V_j.

(3) Initialize the $value(V_{j\in Y})$ values. Since the load of each nominator node is taken as 0 before the election, the initial value of $value(V_{j\in Y})$ can be set as the reciprocal of the approval stake of the candidate validator node, i.e.,

$$value(V_{j\in Y}) = \frac{1}{\mathrm{approval_stake}(V_j)} \qquad \text{(Formula 2)}$$

This sets the initial value of $value(V_{j\in Y})$ to be inversely proportional to the total support weight (approval stake) of each candidate validator node.

(4) Update the $value(V_{j\in Y})$ of the candidate nodes. For each nominator node, iteratively update the $value$ of each candidate validator node that they support by multiplying their weight by the load of the candidate validator node and then dividing by the total weight supporting that candidate validator node, i.e.,

$$value(V_{j\in Y}) = value(V_j) + load(V_j \rightarrow N_i) * stake(V_j \rightarrow N_i)/approval_stake(V_j)$$

(Formula 3)

where $load(V_j \rightarrow N_i)$ represents the load of the nominator node that supports the candidate

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validator node V_j and where $stake(V_j \rightarrow N_i)$ represents the weight (stake) of the nominator node that supports the candidate validator node V_j.

(5) Identify the candidate validator node with the minimum $value(V_{j \in Y})$ as the elected validator node for this round and remove it from the set of candidate validator nodes.

(6) Update the load of the nominator nodes that support the selected validator node by setting their load to the value of the selected candidate node's $value$.

(7) If there are still validator nodes to be elected, return to step 3 and repeat the process.

The above process can be represented as a computer algorithm, as shown in Algorithm 4–1.

Algorithm 1: WNPoS Validator Node Election Algorithm

Input: A list of elections by all nominator nodes;

Output: Elected candidate validator nodes.

load(Ni)={0}

CV_Set={V1,V2,V3,…,Vncv}// *CV_Set represents the set of candidate validator nodes, and ncv represents the number of candidate validator nodes.*

N_Set={N1,N2,N3,…,Nnn }// *N_Set represents the set of nominator nodes, and nn represents the number of nominator nodes approval_stake(V_j)=0.*

value(V_j)=0

for k=1…num_v // *num_v represents the number of validator nodes in the blockchain system.*

for j=1…num_c_v do // *num_c_v represents the number of candidate validator nodes in the blockchain system.*

for i=1…num_n do // *num_c_v represents the number of nominated nodes in the blockchain system.*

if vote(N_i,V_j) then // *vote(N_i,V_j) indicates that the nominated node N_i votes for the candidate verification node V_j.*

approval_stake(V_j)= approval_stake(V_j)+ stake(N_i) // *stake(N_i) represents the weight of N_i node, and approval_stake(V_j) represents the total weight of V_j node.*

end if

end for

for i=1…num_n

if vote(N_i,V_j) then

value(V_j)=value(V_j)+load(N_i)*stake(N_i)/approval_stake(V_j)

end if

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```
        end for
    end for
    selected_validataor=min(value(V)) // Select the candidate validator node with the smallest value and assign it to selected_validator.
    add(selected_validataor,V_Set) // Add the selected verification node to the verification node set V_Set.
  end for
    return V_Set
```

4 Application and analysis of the WNPoS consensus mechanism algorithm

4.1 Application of the WNPoS Consensus Algorithm

Assumptions: There are 6 nomination nodes: N_1, N_2, N_3, N_4, N_5, and N_6;

There are 6 candidate verification nodes: V_1, V_2, V_3, V_4, V_5, and V_6, which compete for 4 verification node seats;

The weights of the N1–N6 nominated nodes are 1–6, represented by stakes;

The support of the nominated node for the candidate verification node is represented by ×;

The support of the nomination node for the candidate verification node is shown in Table 2.

Table 2 Example of Support from Nomination Nodes to Candidate Verification Nodes

Nomination node N	stake	Candidate verification node V					
		V_1	V_2	V_3	V_4	V_5	V_6
N_1	1	×	×			×	
N_2	2	×	×			×	
N_3	3	×					
N_4	4		×	×	×	×	
N_5	5	×		×		×	
N_6	6	×			×		

(1) Calculate the support weight of each candidate verification node according to formula (1):

$$\text{approval_stake}(V_1) = 1 + 2 + 3 + 5 + 6 = 17$$

$$\text{approval_stake}(V_2) = 1 + 2 + 4 = 7$$

$$\text{approval_stake}(V_3) = 4 + 5 = 9$$

$$\text{approval_stake}(V_4) = 4 + 6 = 10$$

$$\text{approval_stake}(V_5) = 1 + 2 + 4 + 5 = 12$$

$$\text{approval_stake}(V_6) = 0$$

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(2) Calculate the initial score $value(V_i)$ of the candidate verification node according to formula 2 (retaining three decimal places):

$$value(V_1) = \frac{1}{17} = 0.059$$

$$value(V_2) = \frac{1}{7} = 0.143$$

$$value(V_3) = \frac{1}{9} = 0.111$$

$$value(V_4) = \frac{1}{10} = 0.1$$

$$value(V_5) = \frac{1}{12} = 0.083$$

$$value(V_6) = N/A$$

(3) In the first round of election, since the load of the nominators is taken as 0, the value of the candidate verifier is the initial value.

The candidate validator with the lowest value in the first round of election is denoted as V_1, with a value of 0.059. Therefore, V_1 is elected as the official validator.

(4) Update supports the load of each nominated node that is selected as a verification node to be $load(N_i)$= 0.059-0=0.059, as indicated in Table 3.

Table 3 Load of each nominated node after the first round of election

Nomination node	**Stake**	Candidate verification node						First round load
		V_1	V_2	V_3	V_4	V_5	V_6	
N_1	1	×	×			×		0.059
N_2	2	×	×			×		0.059
N_3	3	×						0.059
N_4	4		×	×	×	×		
N_5	5	×		×		×		0.059
N_6	6	×			×			0.059

(5) Update the values of the candidate verification nodes

According to formula 3, the calculation yields:

Nominator node N_1 updates the $value$ of V_2 to:

$$value(V_2) = value(V_2) + load(N_1) * stake(N_1)/approval_stake(V_2)$$

$$= 0.143 + 0.059 * \frac{1}{7} = 0.151$$

Nominator node N_2 updates the $value$ of V_2 to:

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$$value(V_2) = 0.151 + 0.059 * \frac{2}{7} = 0.168$$

Nominator node N_4 updates the $value$ of V_2 to:

$$value(V_2) = 0.168 + 0 * \frac{4}{7} = 0.168$$

Therefore, $value(V_2) = 0.168$

Similarly, we can obtain $value(V_3) = 0.143$, $value(V_4) = 0.135$, and $value(V_5) = 0.122$.

$value(V_5)$ is the smallest, so V_5 is selected for the second round.

The load of each nominated node that supports the elected verification node is updated, as shown in Table 4.

(5) Update the value of the candidate verification node

According to formula 3, the calculation is as follows:

The nominating node N_1 updates the $value$ of V_2 to:

$$value(V_2) = value(V_2) + load(N_1) * stake(N_1)/approval_stake(V_2)$$

$$= 0.143 + 0.059 * \frac{1}{7} = 0.151$$

The nominating node N_2 updates the value of V_2 to: $value(V_2) = 0.151 + 0.059 * \frac{2}{7} = 0.168$

The nominating node N_4 updates the value of V_2 to: $value(V_2) = 0.168 + 0 * \frac{4}{7} = 0.168$

Therefore, $value(V_2) = 0.168$

Similarly, we can obtain $value(V_3) = 0.143$, $value(V_4) = 0.135$, and $value(V_5) = 0.122$.

$value(V_5)$ is the smallest, so V_5 is selected in the second round.

The load of each nominated node that supports the elected verification node is updated, as shown in Table 4.

Table 4 Load of each nominating node after the second round of election

Nomination node	**Stake**	Candidate verification node						First round load	Second round load
		V_1	**V_2**	**V_3**	**V_4**	**V_5**	**V_6**		
N_1	1	×	×			×		0.059	0.122

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N_2	2	×	×			×		0.059	0.122
N_3	3	×						0.059	0.059
N_4	4		×	×	×	×			0.122
N_5	5	×		×		×		0.059	0.122
N_6	6	×			×			0.059	0.059

(6) Update the value of the candidate verification node again

$$value(V_i) = value(V_i) + load(N_i) * stake(N_i)/approval_stake(V_i)$$

The nominating node N_1 updates the score value of V_2 to: $value(V_2) = 0.076$;

The nomination node N_2 updates the score value of V_2 to: $value(V_2) = 0.111$;

The nomination node N_4 updates the score of V_2 to: $value(V_2) = 0.181$;

$value(V_2) = 0.181$. Similarly, $value(V_3) = 0.232$ and $value(V_4) = 0.184$.

Therefore, V_2 was elected in the third round. The load of each nominated node that supports the elected verification node is updated, as shown in Table 5.

Table 5 Load of each nominated node after the third round of election

Nomination node	**Stake**	Candidate verification node						First round load	Second round load	Third round load
		V_1	V_2	V_3	V_4	V_5	V_6			
N_1	1	×	×			×		0. 059	0. 122	0. 181
N_2	2	×	×			×		0. 059	0. 122	0. 181
N_3	3	×						0. 059	0. 059	0. 059
N_4	4		×	×	×	×			0. 122	0. 181
N_5	5	×		×		×		0. 059	0. 122	0. 122
N_6	6	×			×			0. 059	0. 059	0. 059

According to the similar process and algorithm of the previous three rounds, we can obtain: $value(V_3) = 0.258$, $value(V_4) = 0.208$. This round V_4 was elected. Therefore, V_1,V_2,V_4 and V_5 obtain seats as official verification nodes. The load of each nomination node after the fourth round of election is shown in Table 6.

Table 6 Load of each nominated node after the fourth round of the election

Nomination node	**Stake**	Candidate verification node						First round load	Second round load	Third round load	forth round load
		V_1	V_2	V_3	V_4	V_5	V_6				

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N_1	1	×	×			×	0.059	0.122	0.181	0.181
N_2	2	×	×			×	0.059	0.122	0.181	0.181
N_3	3	×					0.059	0.059	0.059	0.059
N_4	4		×	×	×	×		0.122	0.181	0.208
N_5	5	×		×		×	0.059	0.122	0.122	0.122
N_6	6	×			×		0.059	0.059	0.059	0.208

4.2 Analysis of the WNPoS Consensus Algorithm

The WNPoS mechanism incentivizes all token holders to engage in the blockchain consensus process as nominated nodes. Nominator nodes will commit tokens to candidate validator nodes. As indicated in the previous section, a higher ranking in the number of token pledges increases the likelihood of a candidate validator node being chosen as a formal validator. Furthermore, nodes that qualify as validator nodes receive equal amounts of rewards, as illustrated in Figure 5.

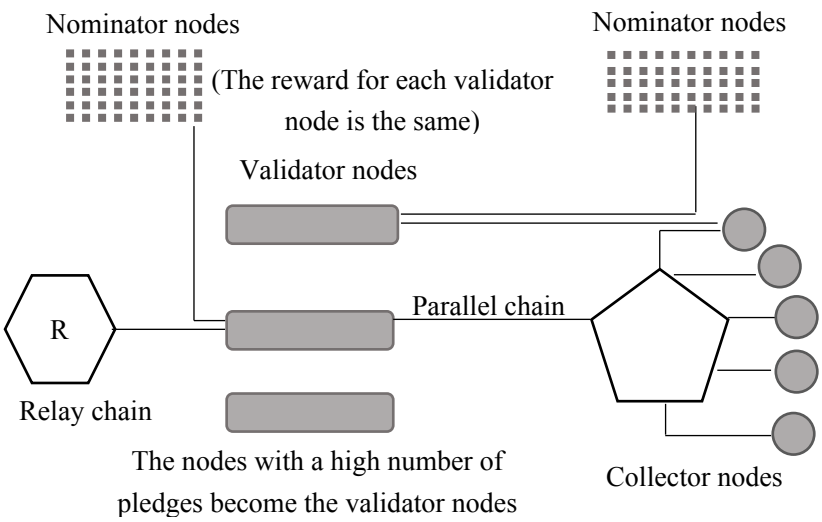

Figure 5 Validator node generation and operation mechanism

In this manner, the unit income received by the nominator node for a validator node with a large number of pledges is relatively low. Consequently, the nominator node is more inclined to pledge tokens to high-quality candidate validator nodes that have fewer total pledges. After numerous rounds of interactions, it can be observed that the total number of tokens staked across all nodes remains fairly consistent, and there are sufficient nominator nodes to provide support. Thus, the WNPoS election process reflects the principles of "fair representation" and "security."

1 Fair Representation

Fair representation implies that the node pool is represented proportionally by the chosen verification nodes, avoiding both overrepresentation and underrepresentation relative to their stacking. This principle of fair representation can be expressed as follows: any nominating node possessing at least 1/n of the total shares ensures that at least one of its trusted verification nodes is

selected.

Among the four figures in Figure 6, Figure 7, Figure 8, and Figure 9, Figures 6 and 7 are considered fairly representative, while Figures 8 and 9 do not meet this criterion. They contravene the principle that "nominating nodes with at least 1/n (where n = 4 in this example) of the total shares have trusted verification nodes elected."

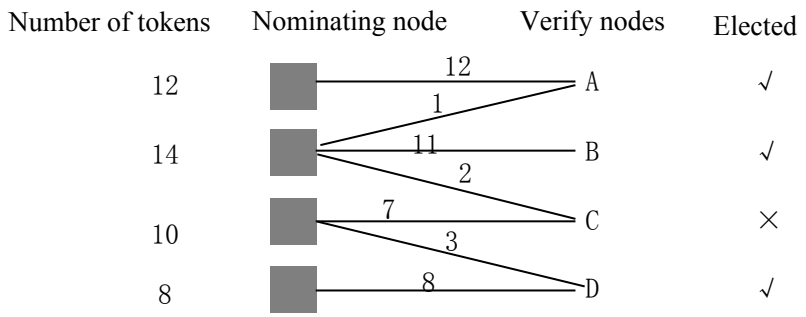

Figure 6 Example 1 of the elected verification nodes demonstrating fair representation

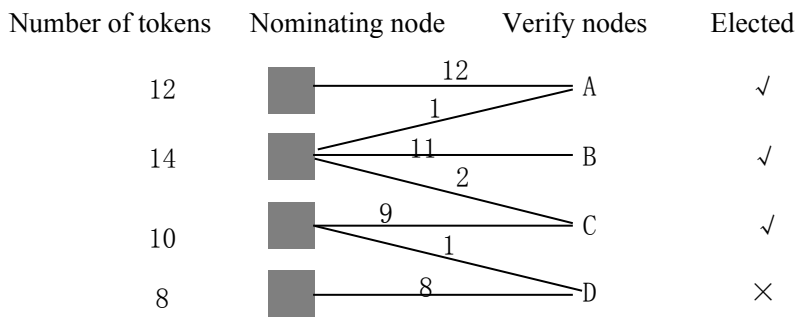

Figure 7 Example 2 of the elected verification nodes demonstrating fair representation

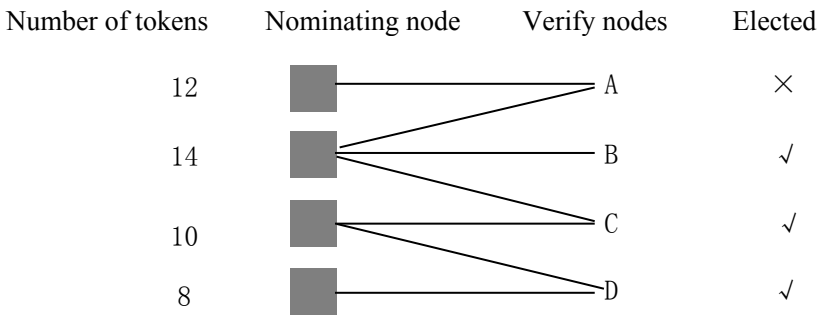

Figure 8 Example 1 of the elected verification nodes not demonstrating fair representation

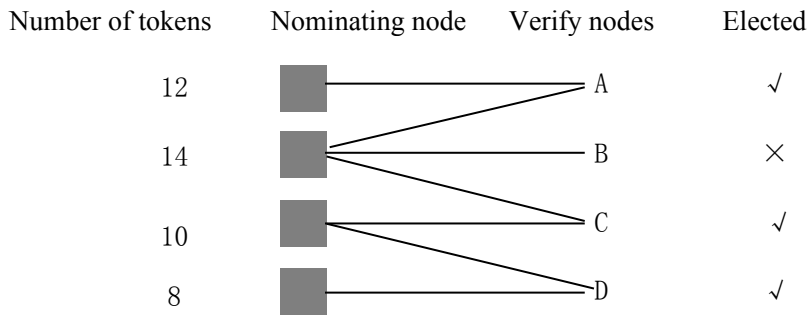

Figure 9 Example 2 of the elected verification nodes not demonstrating fair representation

Under the WNPoS mechanism, scenarios similar to those depicted in Figure 8 or Figure 9 cannot arise because nominating nodes with 12 or 14 tokens, which exceed 1/n (where n = 4 in this example) of the total number of tokens, cannot result in the verification nodes not receiving tokens exceeding those pledged. This holds true regardless of the verification nodes to which tokens are allocated or the amounts pledged.

2 Security

In the WNPoS consensus mechanism, several candidate verification nodes with the highest total stacking amounts are chosen. Therefore, for a candidate verification node to be elected, adequate nomination support must be obtained. As a result, the security level of the election outcome can be defined by the minimum support received from the selected verification nodes. In Figure 10, the minimum support amount is 11, signifying that the election has attained a security level of 11. This minimum number of nominations is essential for maintaining the security of the verification node.

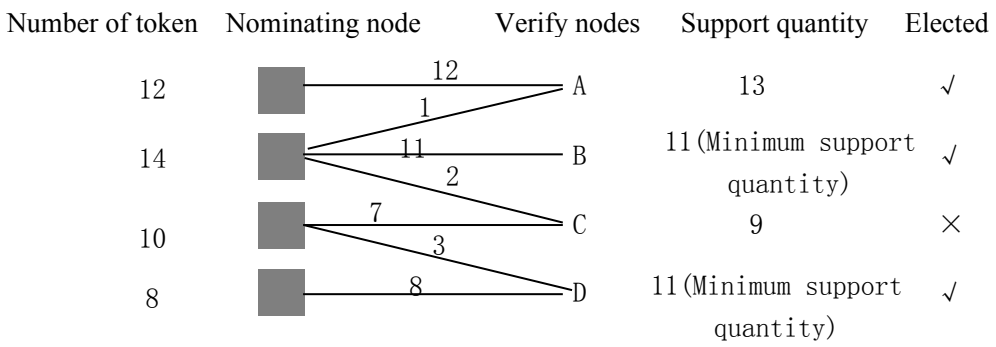

Figure 10 Security level of the verification nodes

5 Conclusion

The consensus mechanism is an essential aspect of the blockchain system, contributing to its stable functioning. Given the characteristics of blockchain cross-chain systems that involve multiple participants and complex operations, a WNPoS consensus algorithm is introduced in this article, describing its principles and attributes. This algorithm ensures that candidate verification nodes with

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higher rankings in terms of token pledges are more likely to be chosen as official verification nodes; however, all nodes receive the same reward, which enhances "fair representation" and "security" in the election process.

The the first 2 lines of this paragraph seem a bit exaggerative. The last 2 lines seems as a part of a review report, not a conclusion!

This study aids in advancing the innovation of consensus mechanisms and enhancing blockchain governance theory while facilitating the development of the sharing economy and expanding the application scenarios of blockchain technology. Nonetheless, several limitations exist within this study. First, an in-depth comparison of the WNPoS consensus algorithm's performance with that of other algorithms has not been conducted. Second, the full implementation of the WNPoS consensus algorithm has yet to be achieved. Future research may further investigate these aspects.

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Conflict of interest

The author claims no conflicts of interest, financial or otherwise.

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